gamma rays emitted.

**6** **(a)** Magnetic flux linkage of the coil = number of turns x magnetic flux density x area of the coil perpendicular to the magnetic field

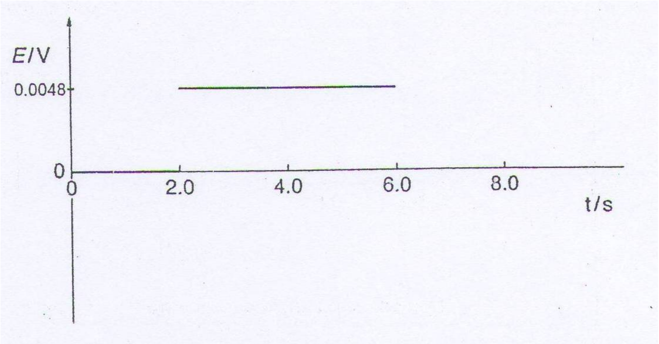
$=15(0.018)(0.30\times 0.24)$

$=0.019$ Wb turns

**(b)** **(i)** The change in magnetic flux density between 2.0 s and 6.0 s results in a rate of change of magnetic flux linkage. According to Faraday's law of electromagnetic induction, this will result in an induced e.m.f. which is directly proportional to the rate of change of magnetic flux linkage.

**(ii)** $E=\dfrac{0.019}{4.0}=4.8\times 10^{-3}\,\text{V}$

**(iii)**




**7** **(a)** In an ideal gas equation of state, the product of the pressure $P$ and volume $V$ of an